

ANIRUDH RAGHAVAN

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anirudh-r2911.github.io/Portfolio | github.com/ANIRUDH-R2911

EDUCATION

Purdue University, West Lafayette, IN – MS in Autonomy (GPA 3.83/4.0) Aug 2024 – May 2026

Coursework: Deep Learning for Computer Vision, Embedded Systems, Autonomous Systems, Artificial Intelligence

Vellore Institute of Technology, Chennai, India – B. Tech, Electronics and Communication Engineering (GPA 8.11/10) 2020 – 2024

TECHNICAL SKILLS

Languages & Frameworks: Python, C++, C, PyTorch, TensorFlow, OpenCV, MATLAB, NumPy, Pandas, Scikit-learn

Computer Vision & ML: Object Detection (YOLO), Semantic & Video Object Segmentation (U-Net, SegFormer), CNNs, Transformers, Model Fine-Tuning, Transfer Learning, Sensor Fusion, Gesture Recognition

Robotics & Embedded: ROS2, Gazebo, RViz, NVIDIA Jetson Nano, STM32, ESP32, Arduino, Real-Time Processing, IoT

Tools & Platforms: Git, Linux (Ubuntu), Firebase, Node-RED, IBM Cloud

EXPERIENCE

Graduate Research Project (Faculty-Advised) – Purdue University, West Lafayette, IN Aug 2025 – Dec 2025

- Built an end-to-end Python benchmarking framework evaluating YOLOv8, YOLOv9, YOLO12 and Autoware's YOLOX detector, converting Waymo Open Dataset 3D labels into standardized 2D bounding boxes across a unified training and evaluation pipeline.
- Fine-tuned YOLOv8 to 65% mAP@0.5 overall and 85% mAP@0.5 with 86.8% recall on pedestrians, more than doubling Autoware's YOLOX baseline of 39% mAP@0.5 on the safety-critical Person category.

IoT Externship (Remote) – SmartInternz, Hyderabad, India May 2023 – Jul 2023

- Architected a cloud-connected smart parking system using ESP32, integrating ultrasonic sensors for occupancy detection and PIR sensor and servo motor for automated gate control.
- Engineered an MQTT-based data pipeline in Node-RED to synchronize hardware status with IBM Cloud, achieving 98% detection accuracy and sub-second update latency, reducing false availability reports by 85% in Wokwi-simulated environments.

ACADEMIC PROJECTS

Structured Slot Aggregation for Cross-Modal Video Object Segmentation | PyTorch, Transformers, SegFormer | *Fall 2025* | [GitHub](#)

- Designed a dual-stream segmentation framework using guided slot attention to maintain consistent object-centric representations across video frames.
- Resolved temporal instability in high-motion scenes by fusing RAFT-based optical flow with SegFormer visual features, keeping temporal mask drift below 5% under occlusion and rapid camera motion.

Object Detection & Semantic Segmentation on Road Scenes | Python, YOLOv8, U-Net | *Spring 2025* | [GitHub](#)

- Trained and evaluated YOLOv8 and U-Net on identically annotated urban road scenes from the Waymo Open Dataset to directly compare detection and segmentation approaches.
- Benchmarked YOLOv8 against U-Net (79% mean IoU), quantifying the real-time detection vs. pixel-level segmentation trade-off for autonomous-driving perception stacks.

Weather-Invariant Object Detection | Python, YOLOv8, OpenCV, Data Augmentation | *Fall 2024* | [GitHub](#)

- Developed a two-stage adaptive perception pipeline that integrates a CNN-based weather classifier to trigger specific image preprocessing (e.g., dehazing, gamma correction) before inference.
- Prioritized edge-computing efficiency by avoiding full model retraining and improved robustness in adverse conditions, resulting in a 27% reduction in false negatives and 93.1% mAP on custom test splits.

PUBLICATIONS

Anirudh Raghavan, Y. Ramesh, H. Shyam, B. Hency V., M. N. Halgamuge - "Enhancing Earthquake Response with AI: Real-Time Human Detection Using Deep Learning and Robotic Systems," *Results in Engineering (Elsevier, Q1, Impact Factor 7.9)*, 2026. | [Link](#)

Abhishek Sebastian, R. Pragna, K. Vishal Vythianathan, Dasaraju Sohan Sai, U. Shiva Sri Hari Al, R. Anirudh, Apurv Choudhary - "Design of Rubble Analyzer Probe Using ML for Earthquake," *AIP Conference Proceedings 2946(1): 040003*, 2023. | [Link](#)